

Parasitic Capacitance in Josephson Junction Arrays

Origins and Behavior

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Outline

1. Fluxonium Junction Array

- Motivation
- Until now: Dolan Bridge Geometry & Constraints

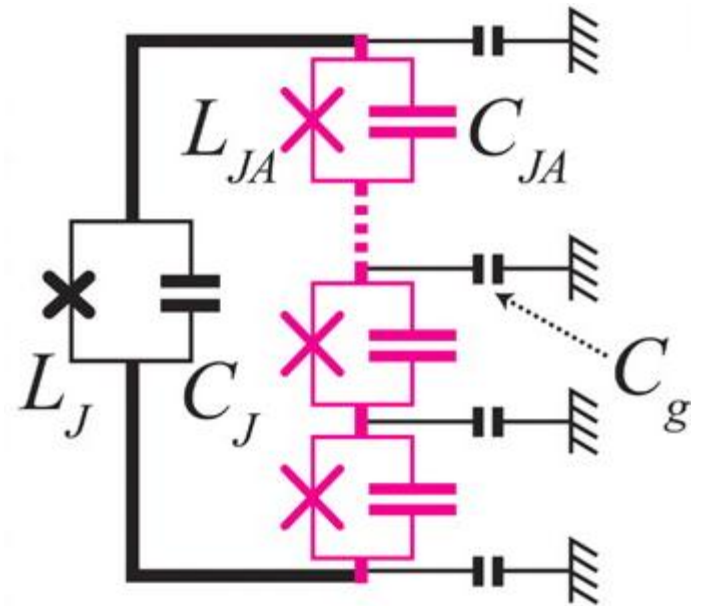
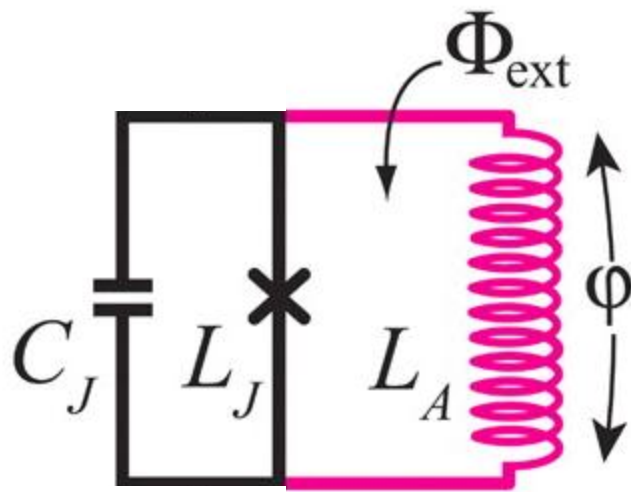
2. Theory & Experiment v.s. Simulation

- Design Study Conclusion & Simple Model
- Resolution: Bridge Free (Island Pop)
- Bridge Free Technique – Skinny & Long

3. In Progress:

- Q Measurement Simulation & Chip Import
- Ground Up Mathematica Assisted Exploration of Fluxonium
 - Aim Super-Fluxonium & Quantum Feedback on λ Energy Structure

Series Junction Array in Fluxonium



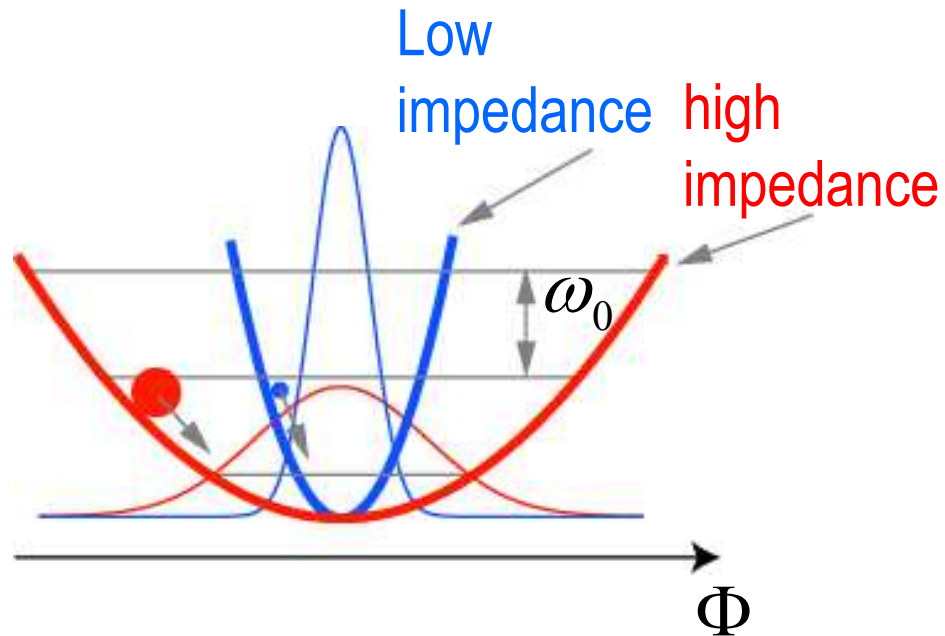
No superconducting islands,
No offset charges!

0.5 2.5

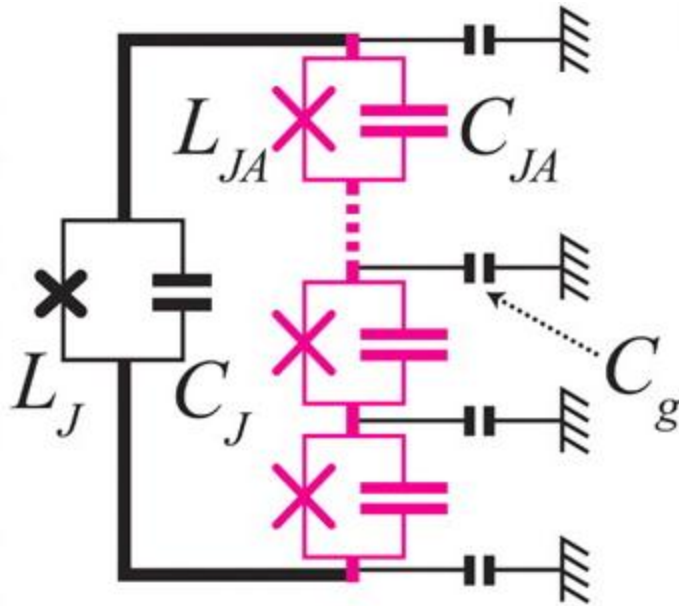
Fx

9.0

Impedance Control ZPF



Capacitance to Ground

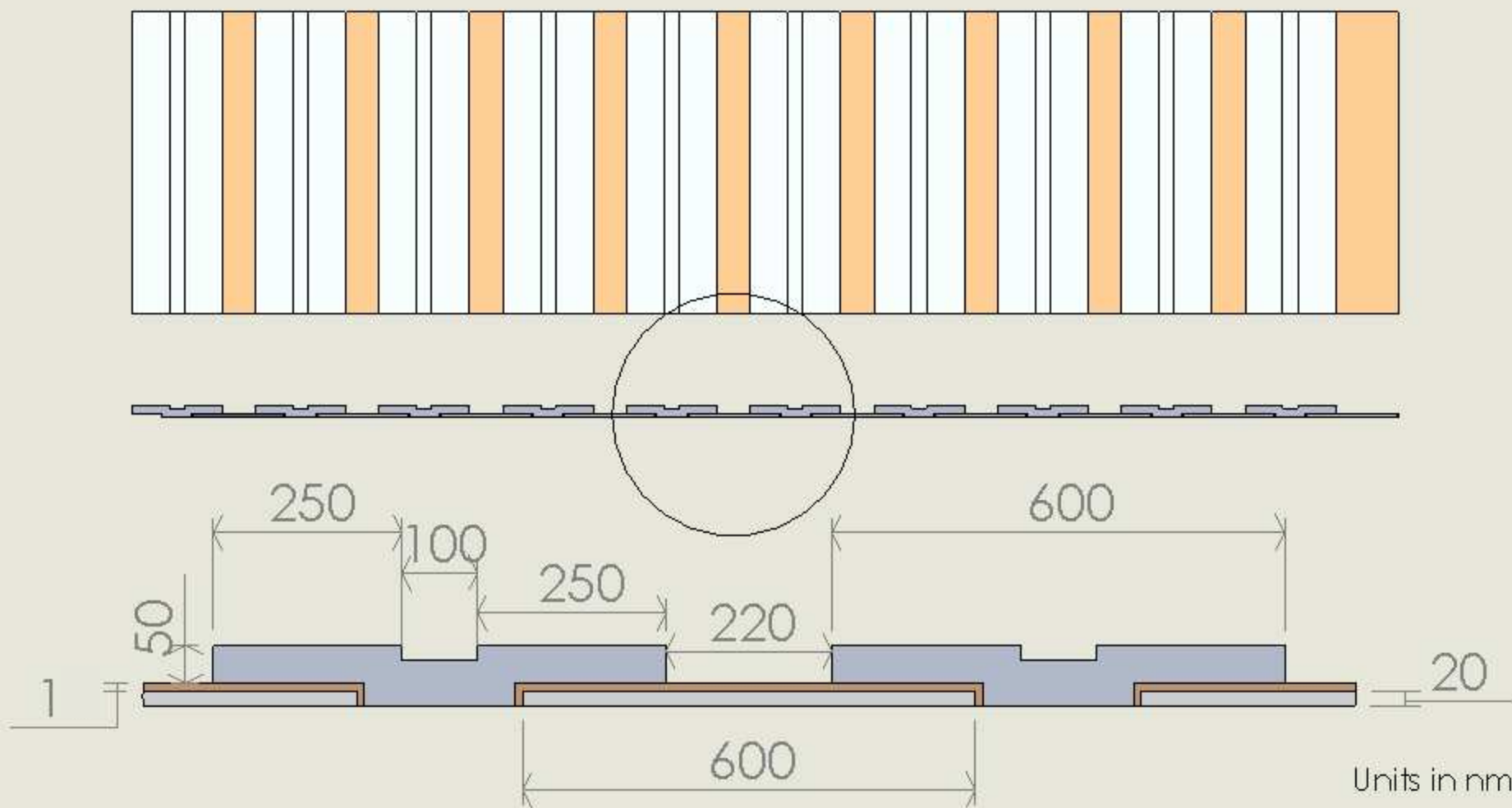


$$\frac{C_{JA}}{N} > N C_g$$

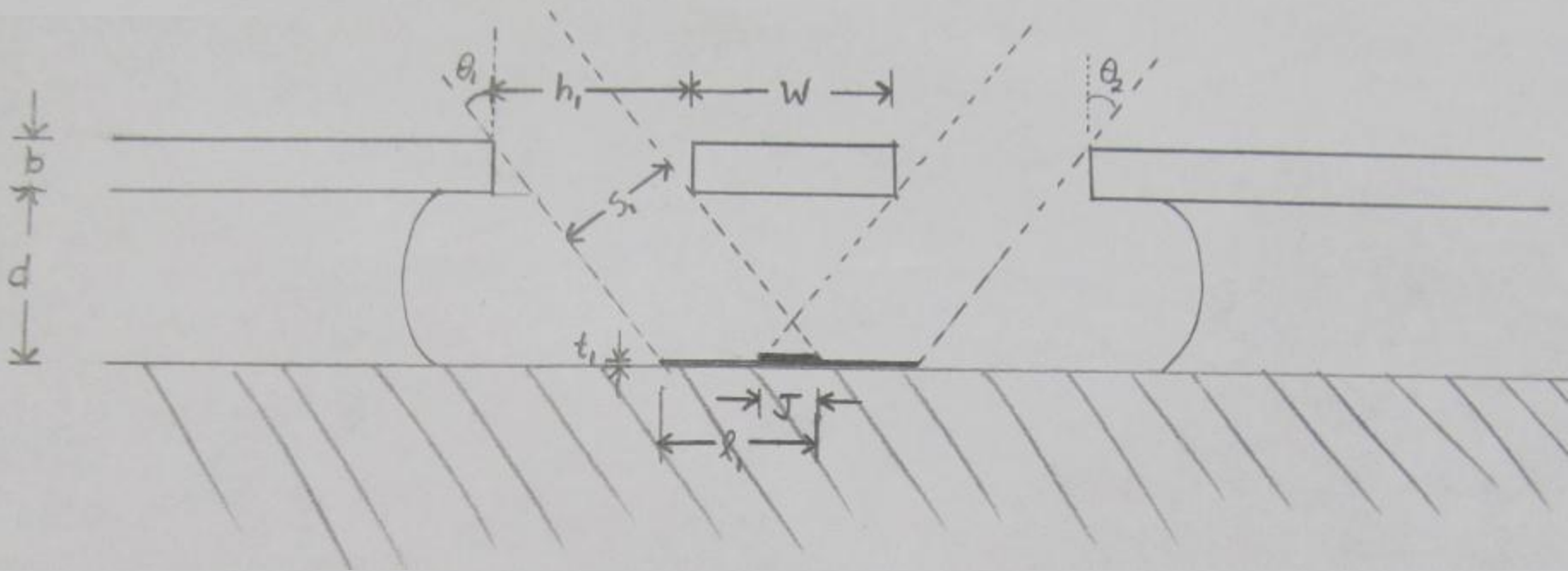
$$N_{\max} \approx \sqrt{C_{JA} / C_g} \approx \sqrt{10^3}$$

2. Fluxonium Junction Array

Junction Array Geometry

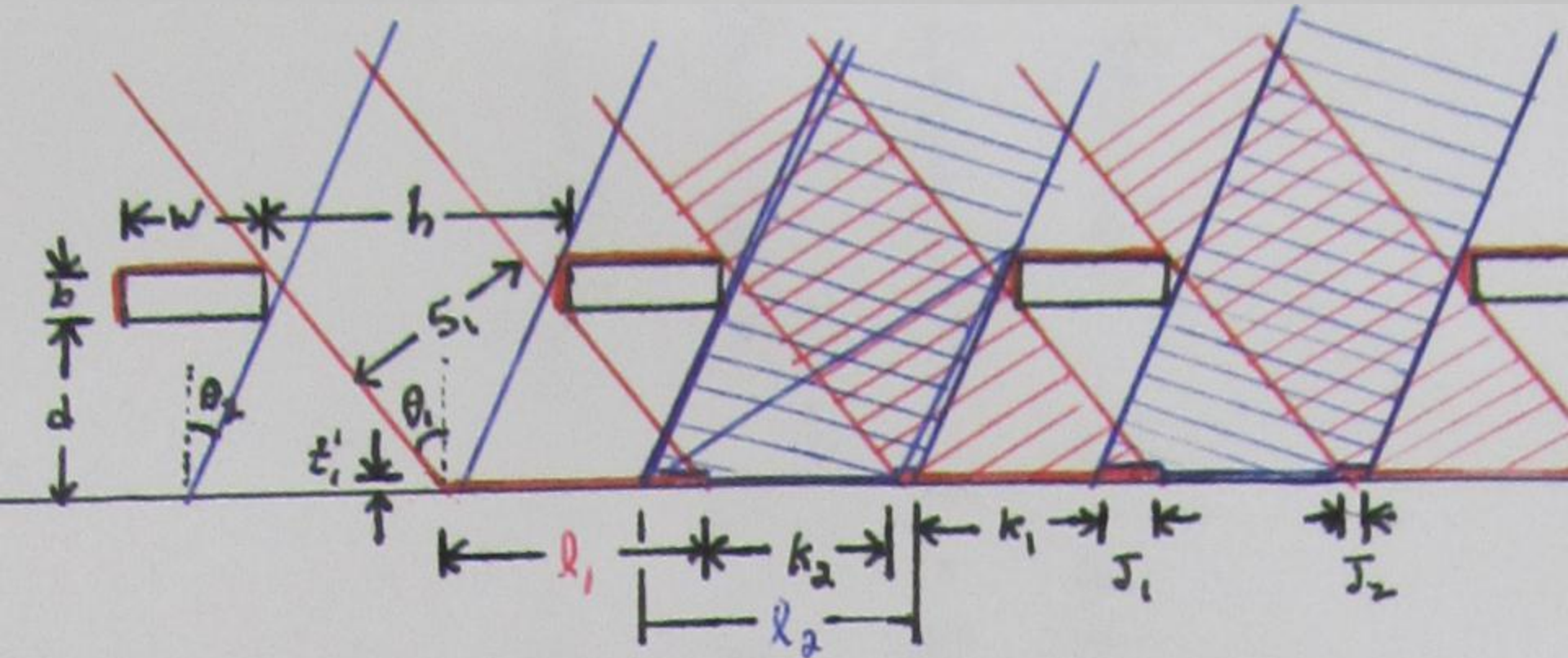


Dolan Bridge Technique



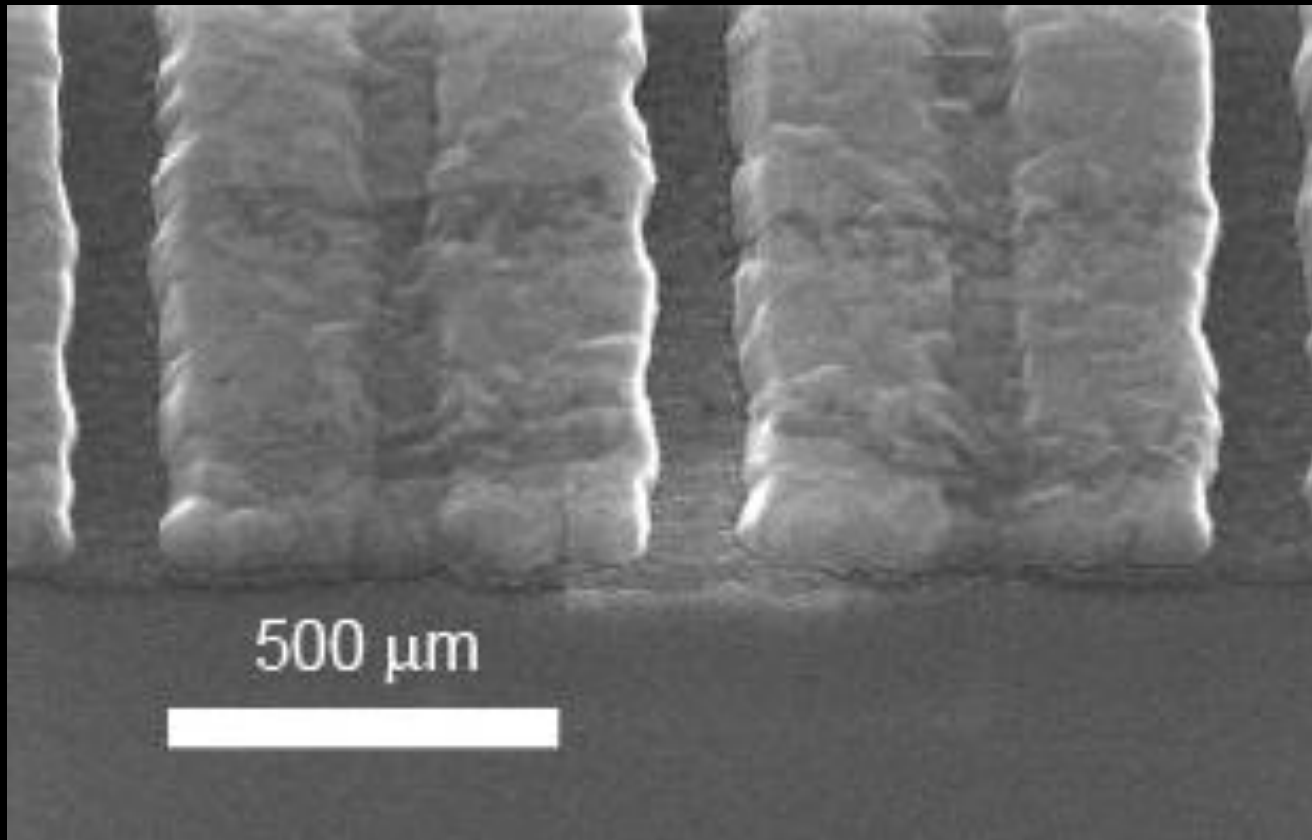
Dolan Bridge

Evaporation 1
Evaporation 2



Nicholas Masluk

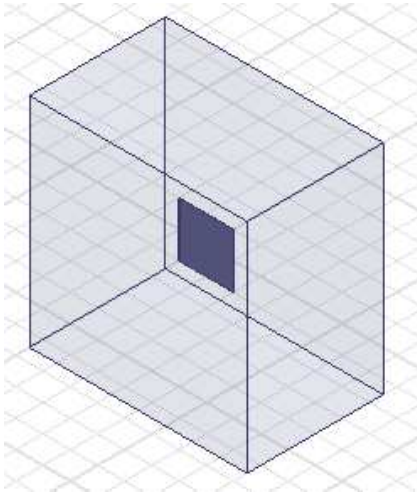
Reality of Junction Array



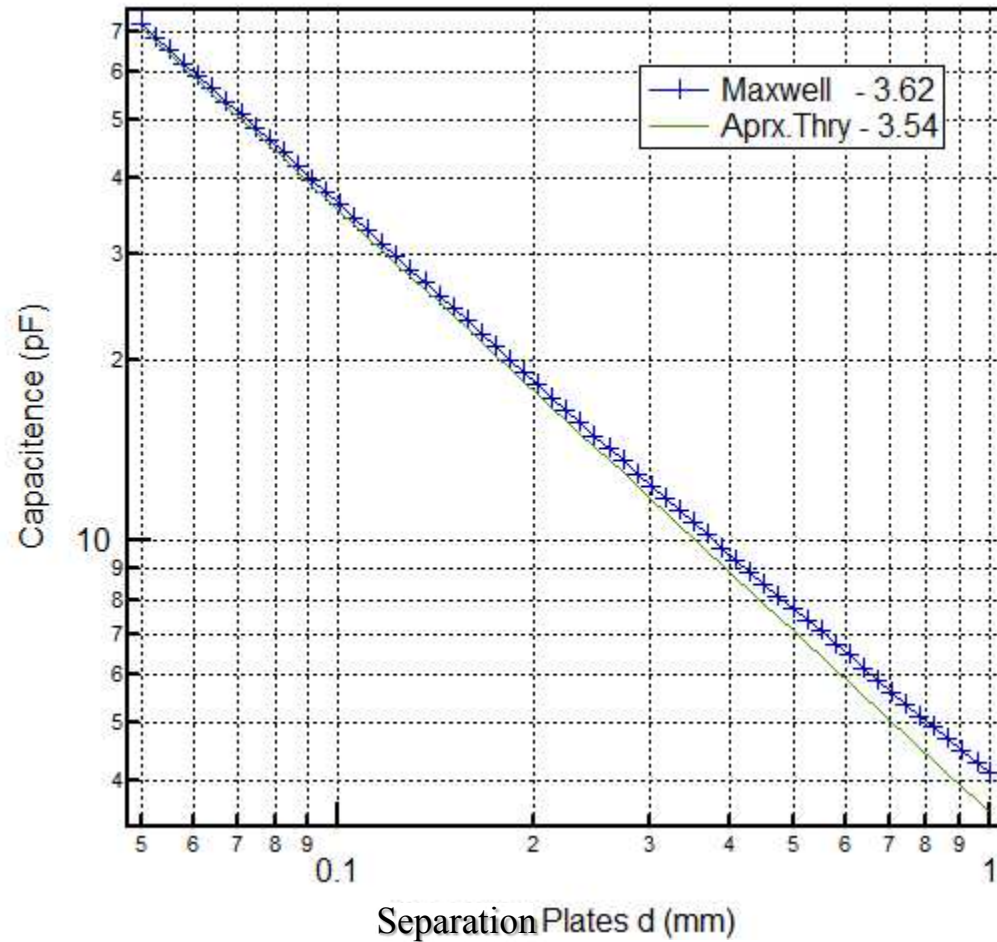
Simulation

Simulation v. Theory

Parallel Plate Capacitor

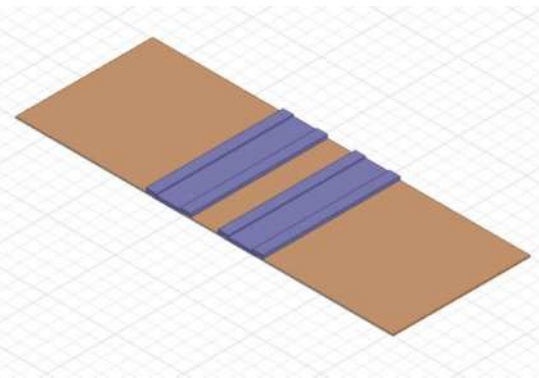


Area = 20 x 20 mm



Simulation v. Empirical Data

Junction Capacitance v. Junction Area

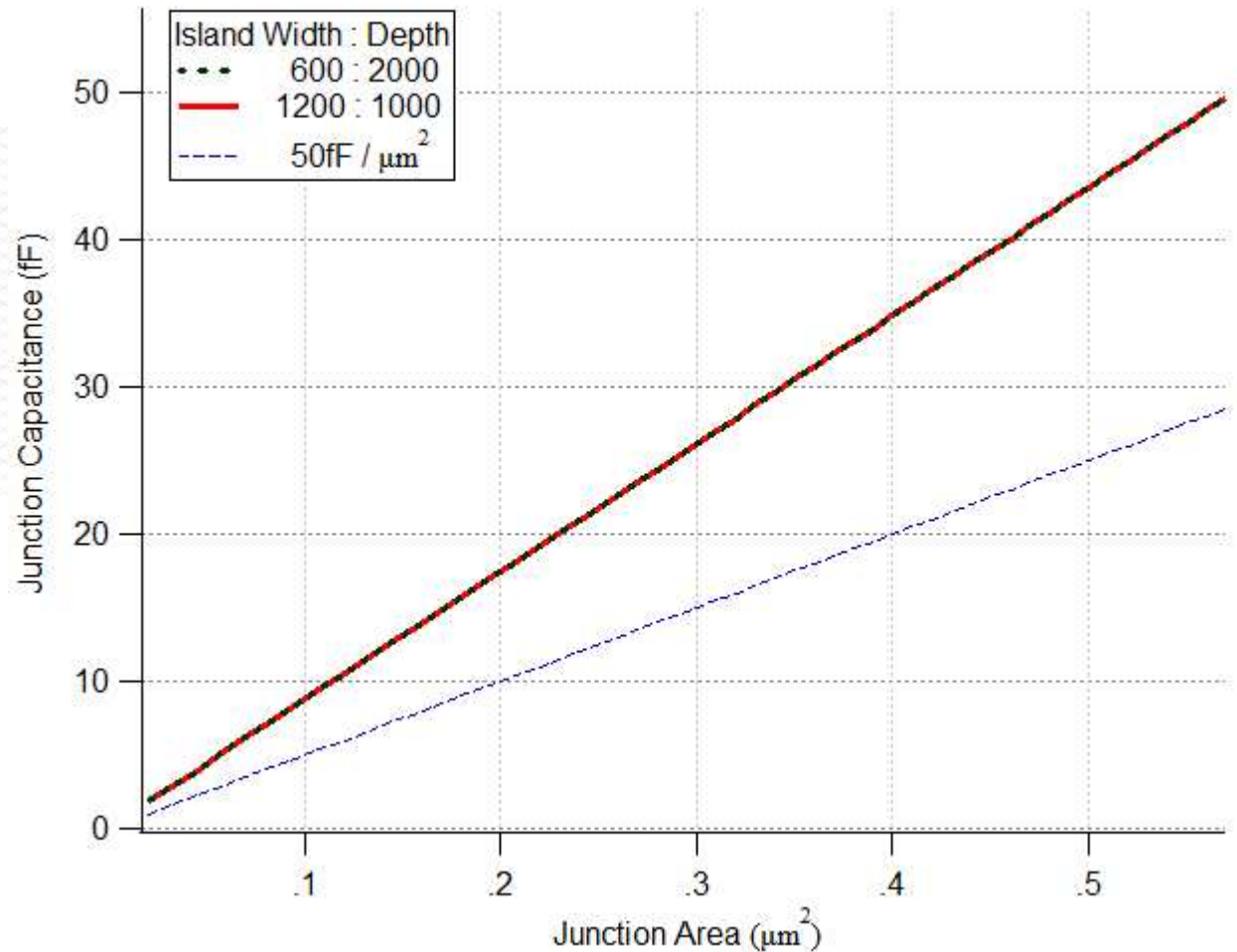


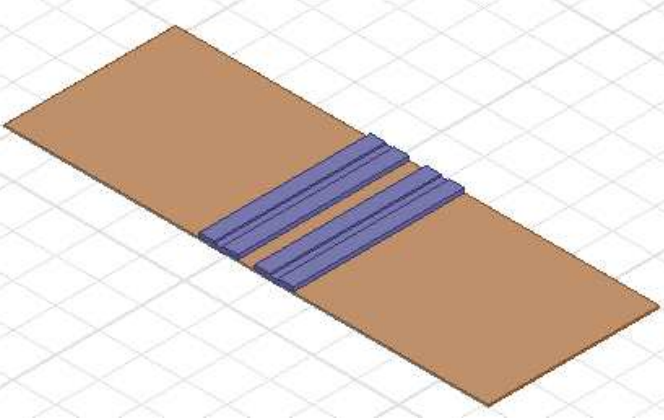
Simulated junction
capacitance per
junction area:

$$\sim 86 \text{ fF} / (\mu\text{m})^2$$

Empirically:

$$\sim 50 \text{ fF} / (\mu\text{m})^2$$





Dolan Bridge Geometry

Vary Island Separation by Varying
Island Width

$d = 10\text{nm}$



$d = 100\text{nm}$



$d = 200\text{nm}$

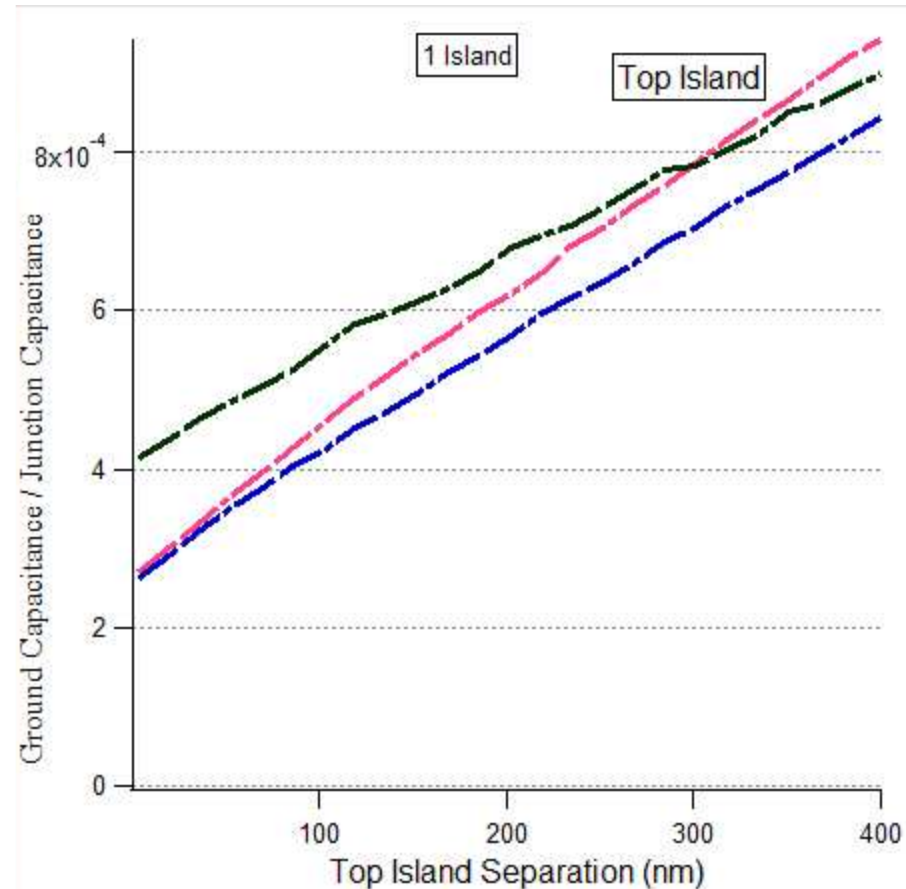
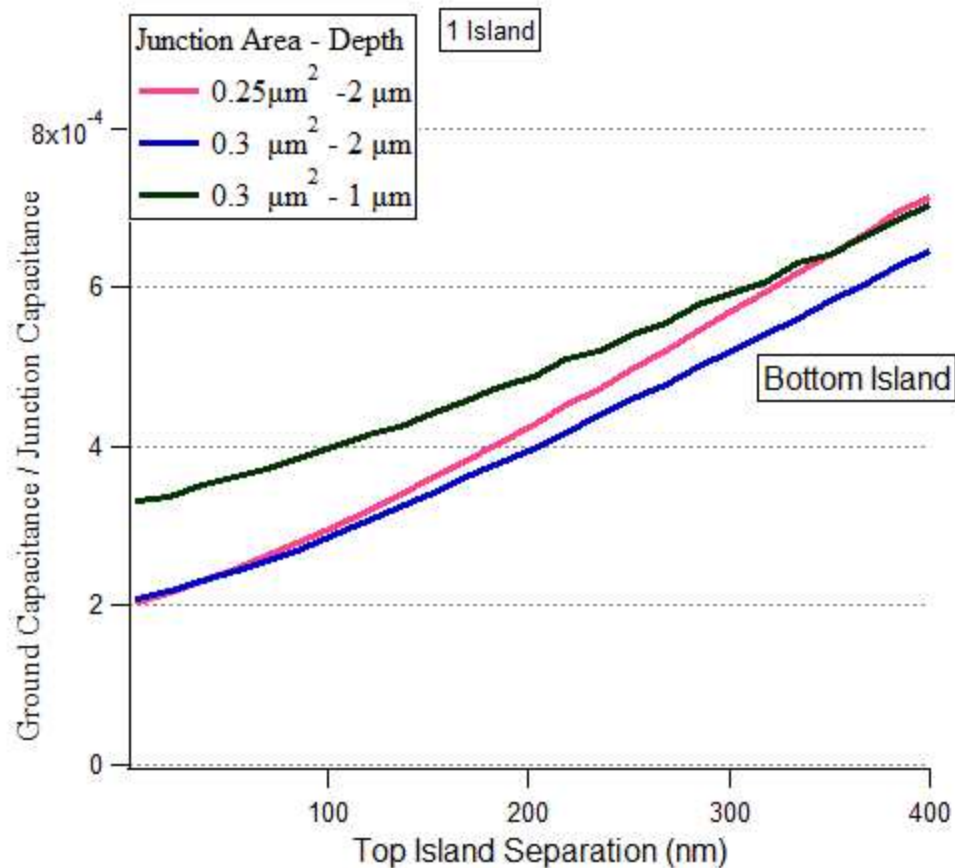


$d = 300\text{nm}$



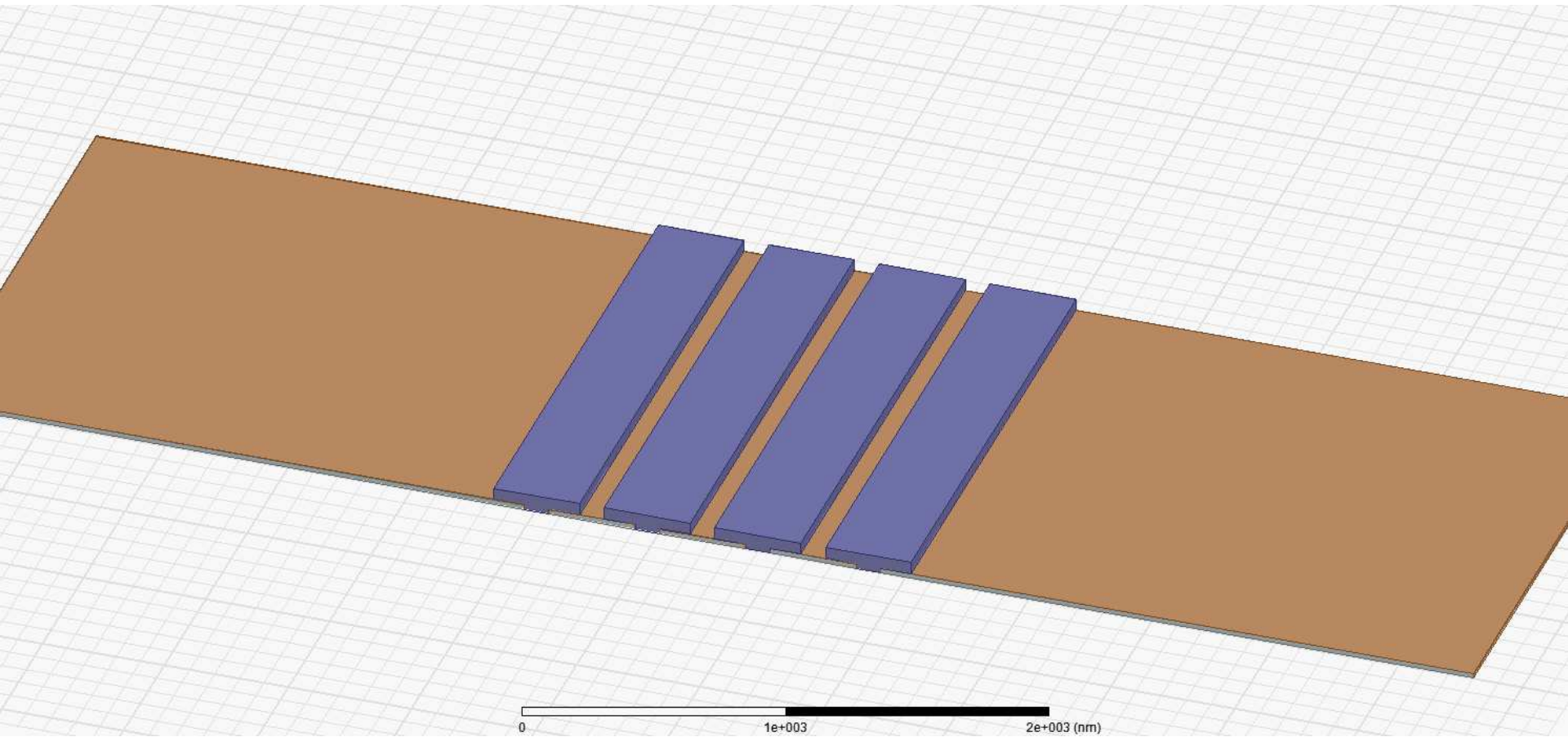
Dolan Bridge Geometry

Vary Island Separation by Varying Island Width

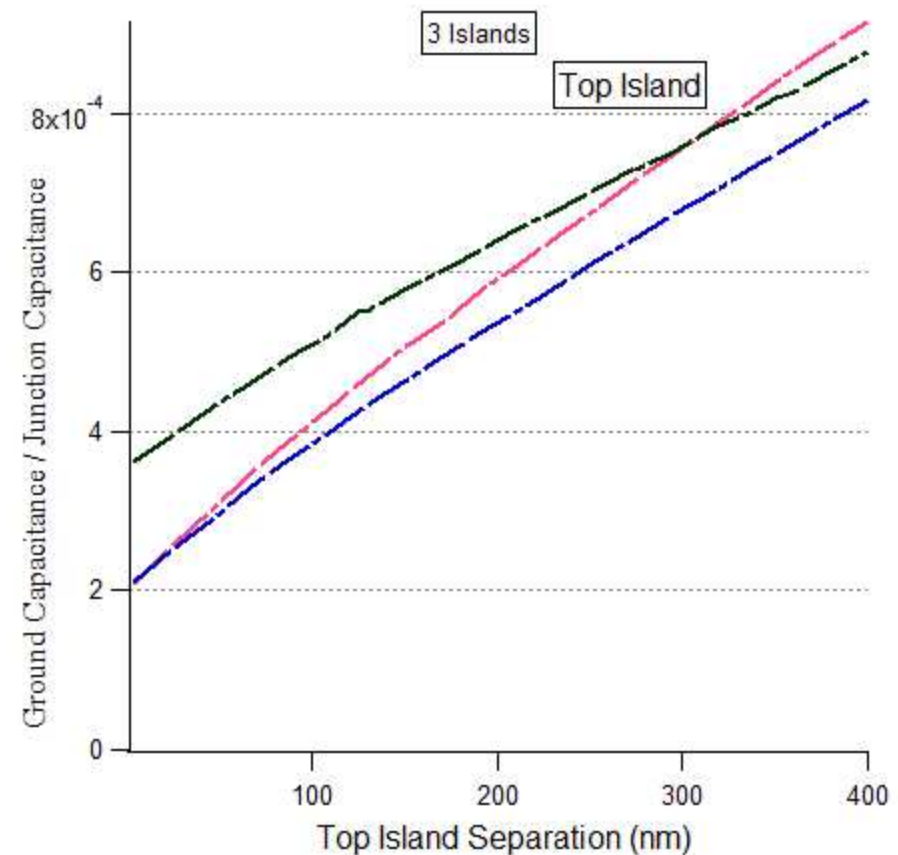
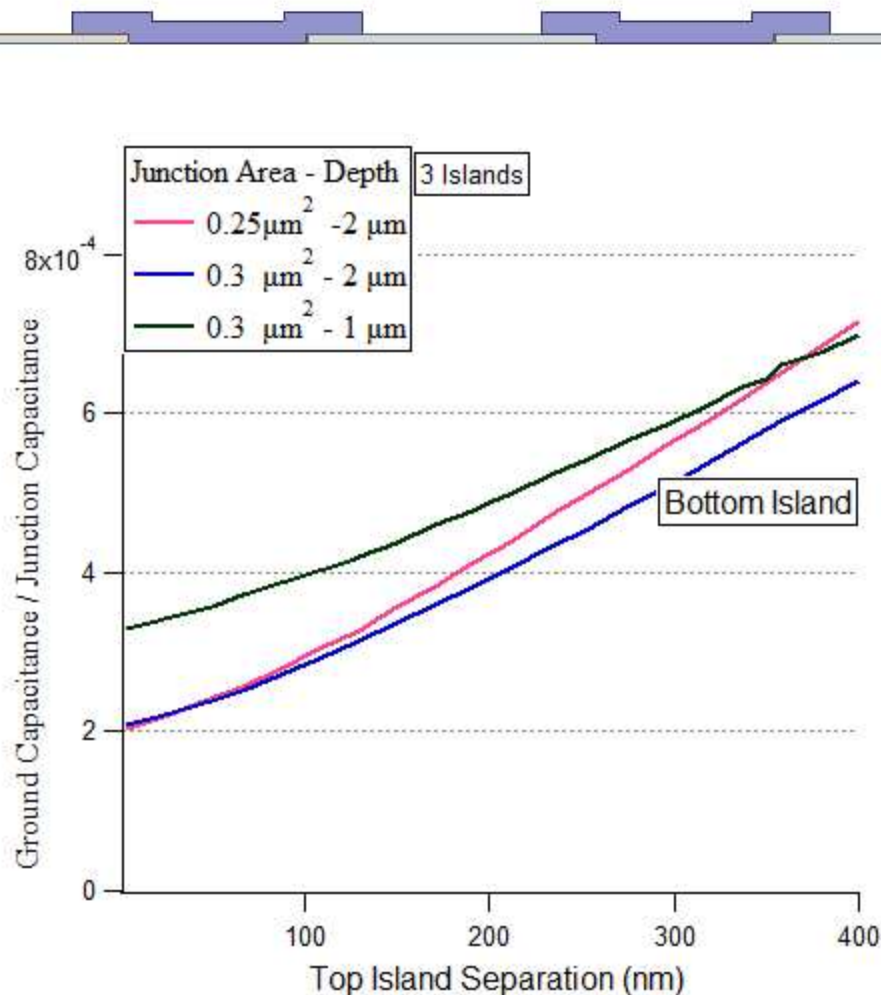


Robustness of Results

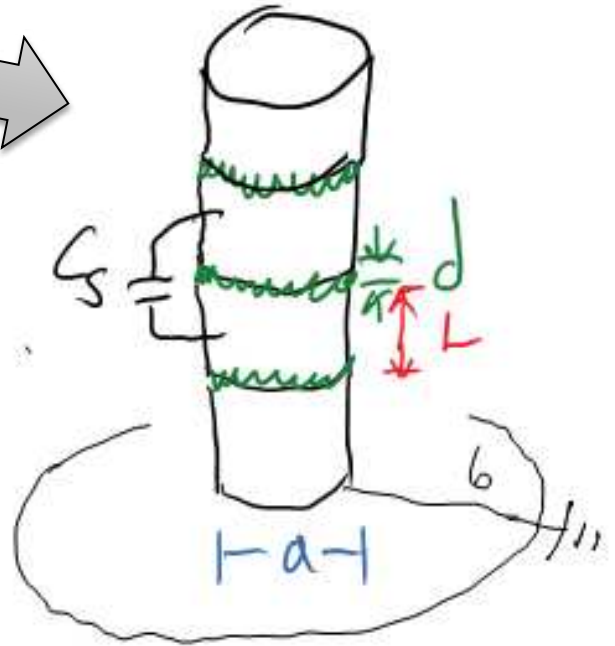
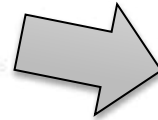
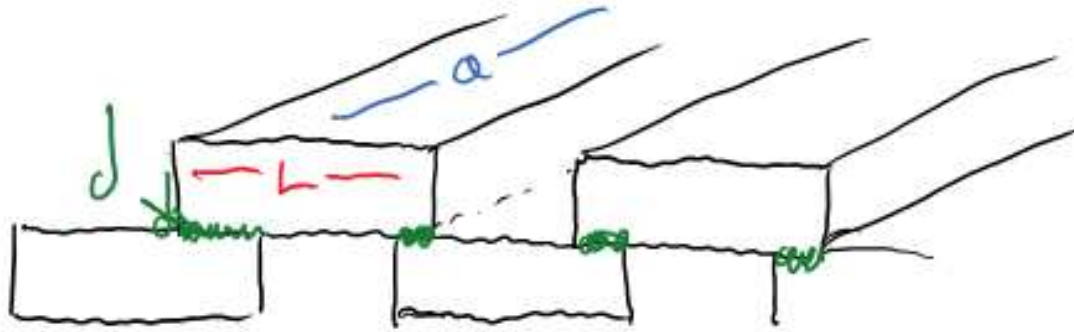
3 v. 1 Middle Islands



Dolan Bridge Geometry



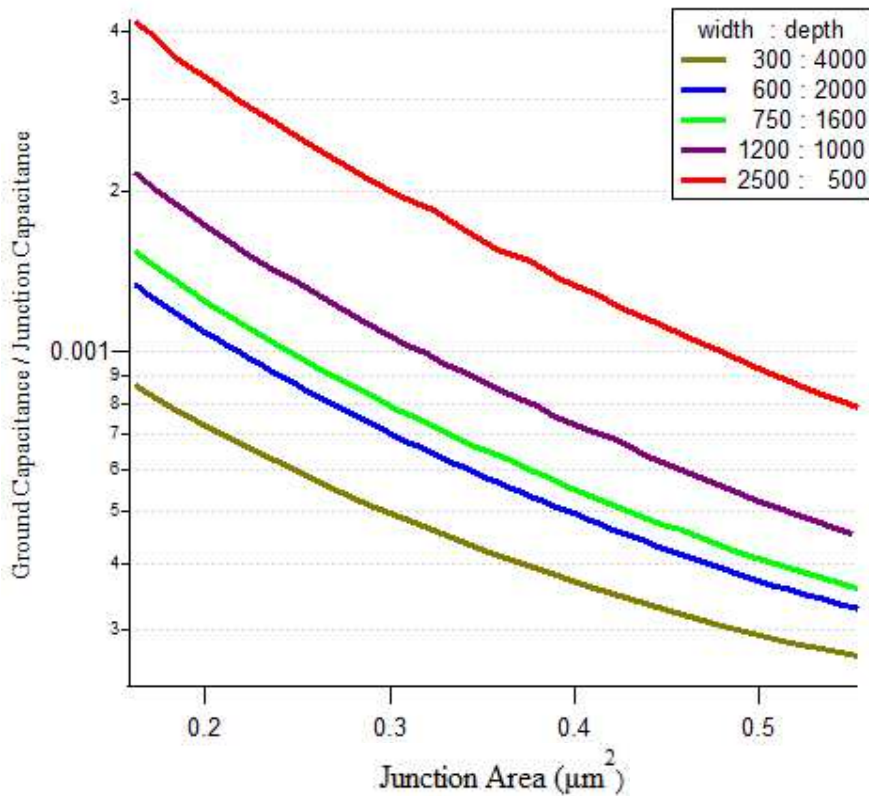
Model



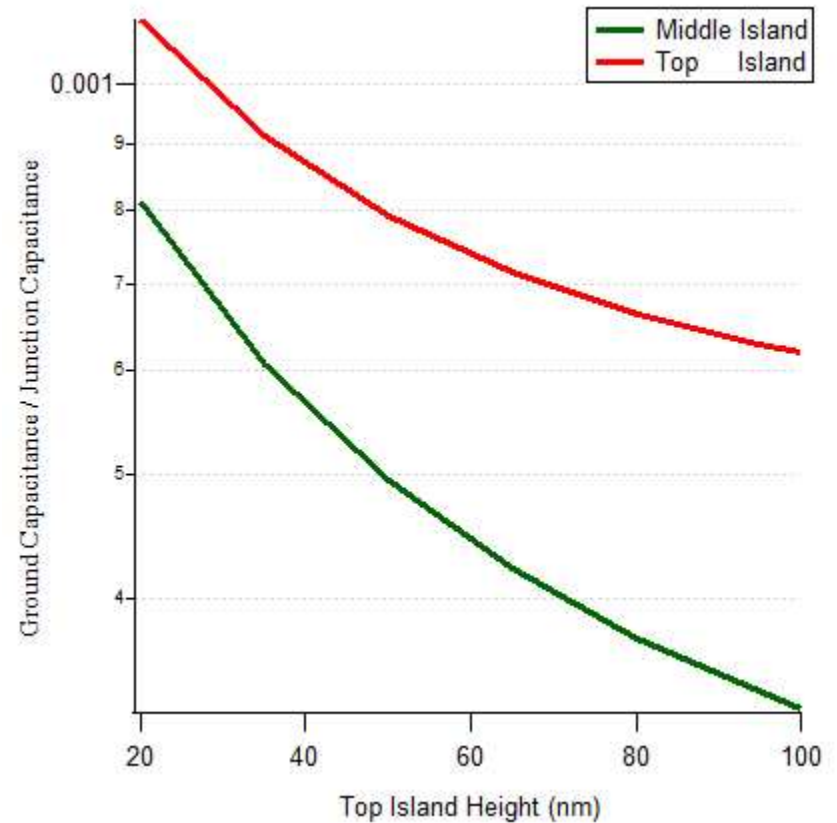
$$\frac{C_g}{C_{ss}} \sim \#L + \#$$

Other Geometrical Variations

Vary Junction Area, Fix Island Size



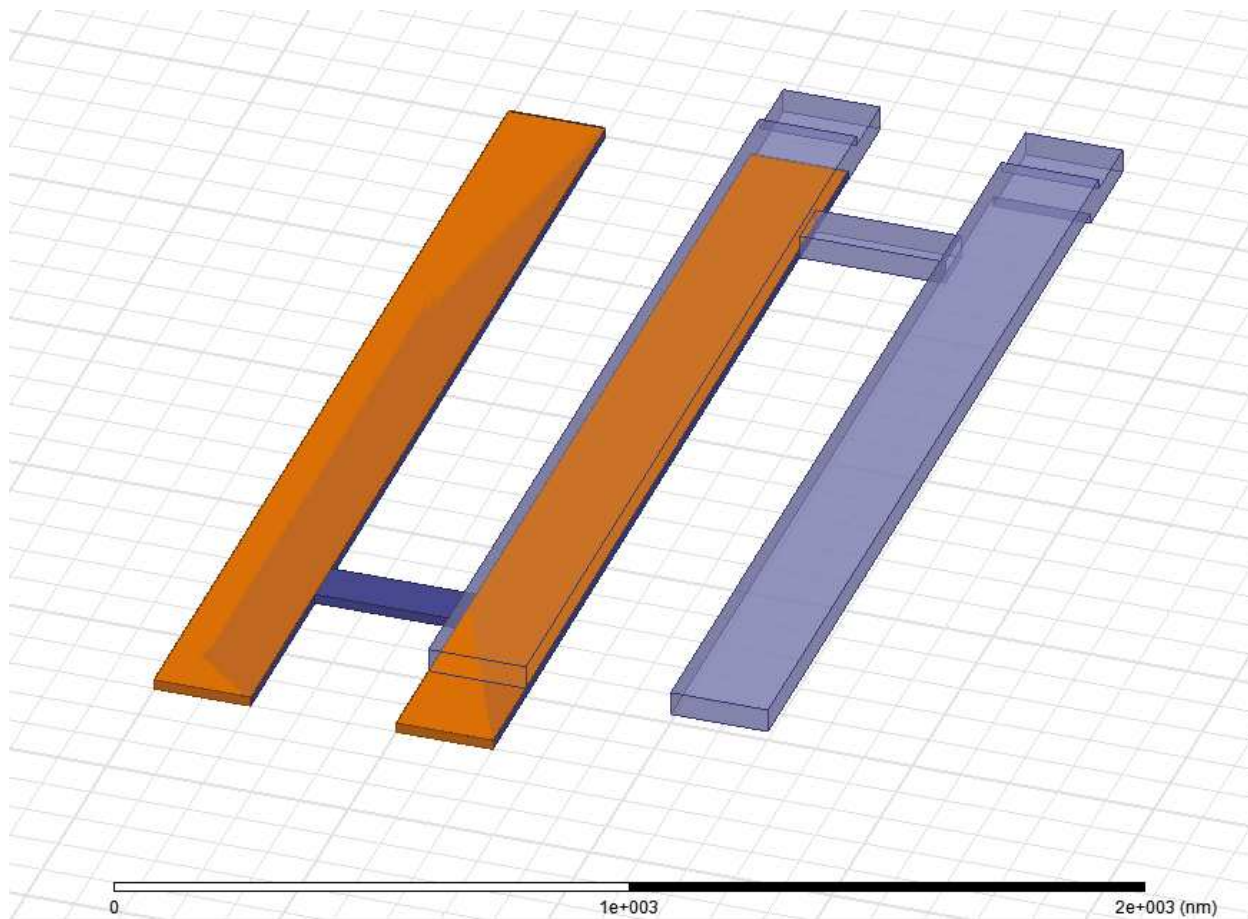
Vary Top Island Height

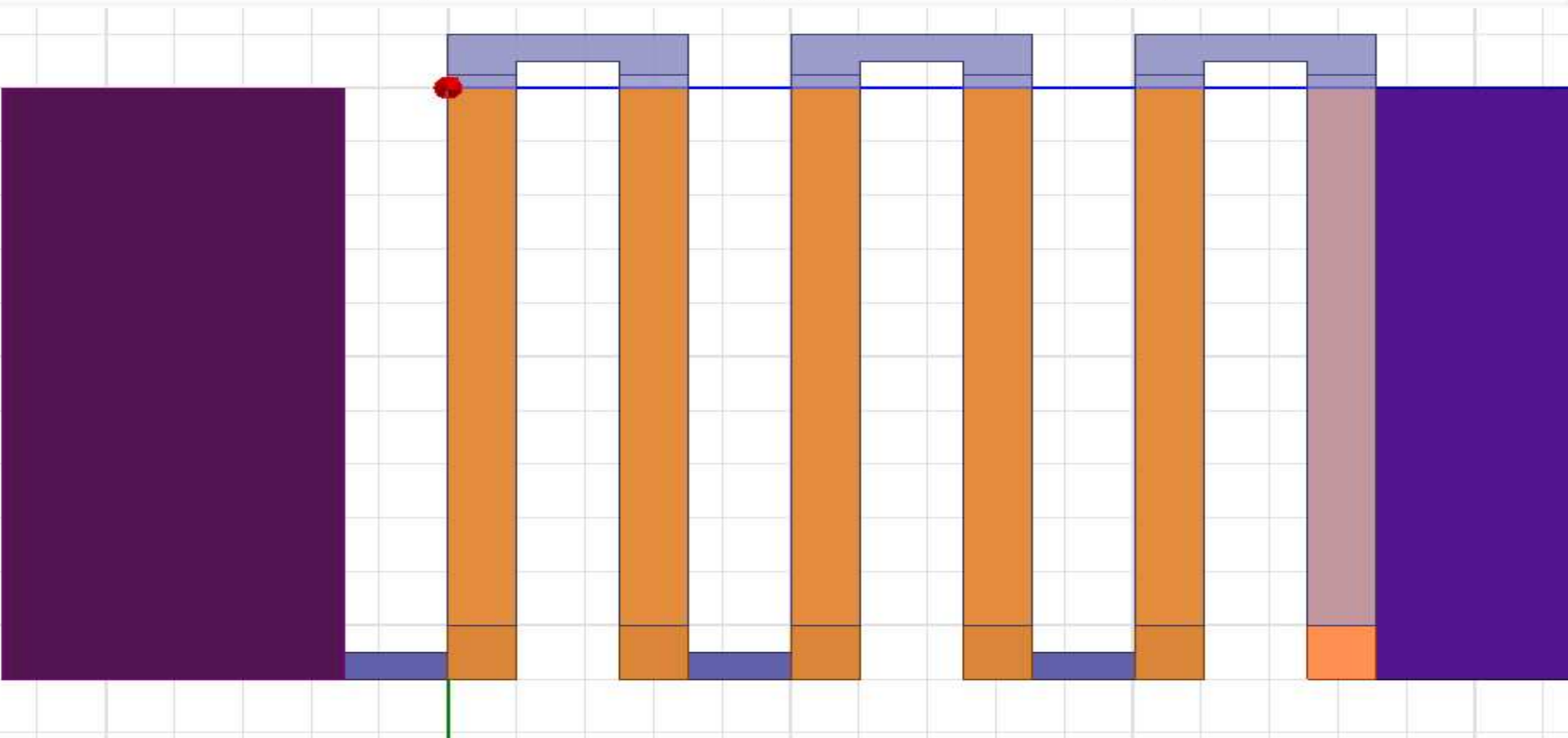
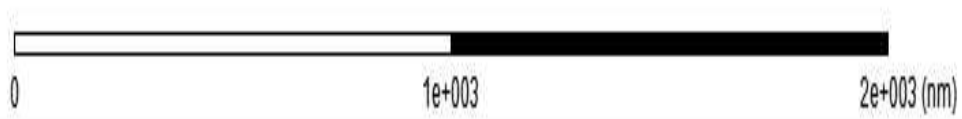


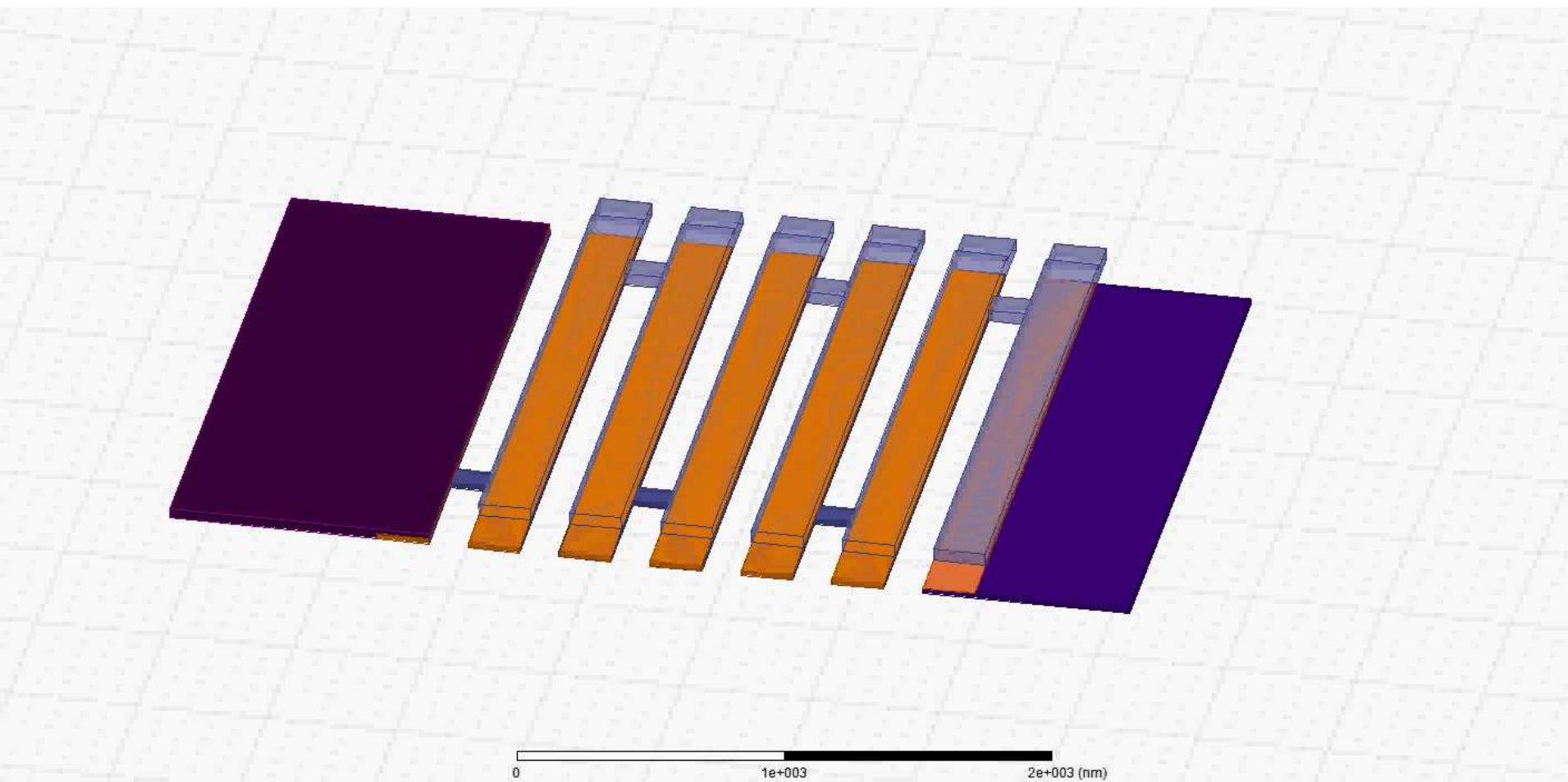
Dolan Bridge Conclusions

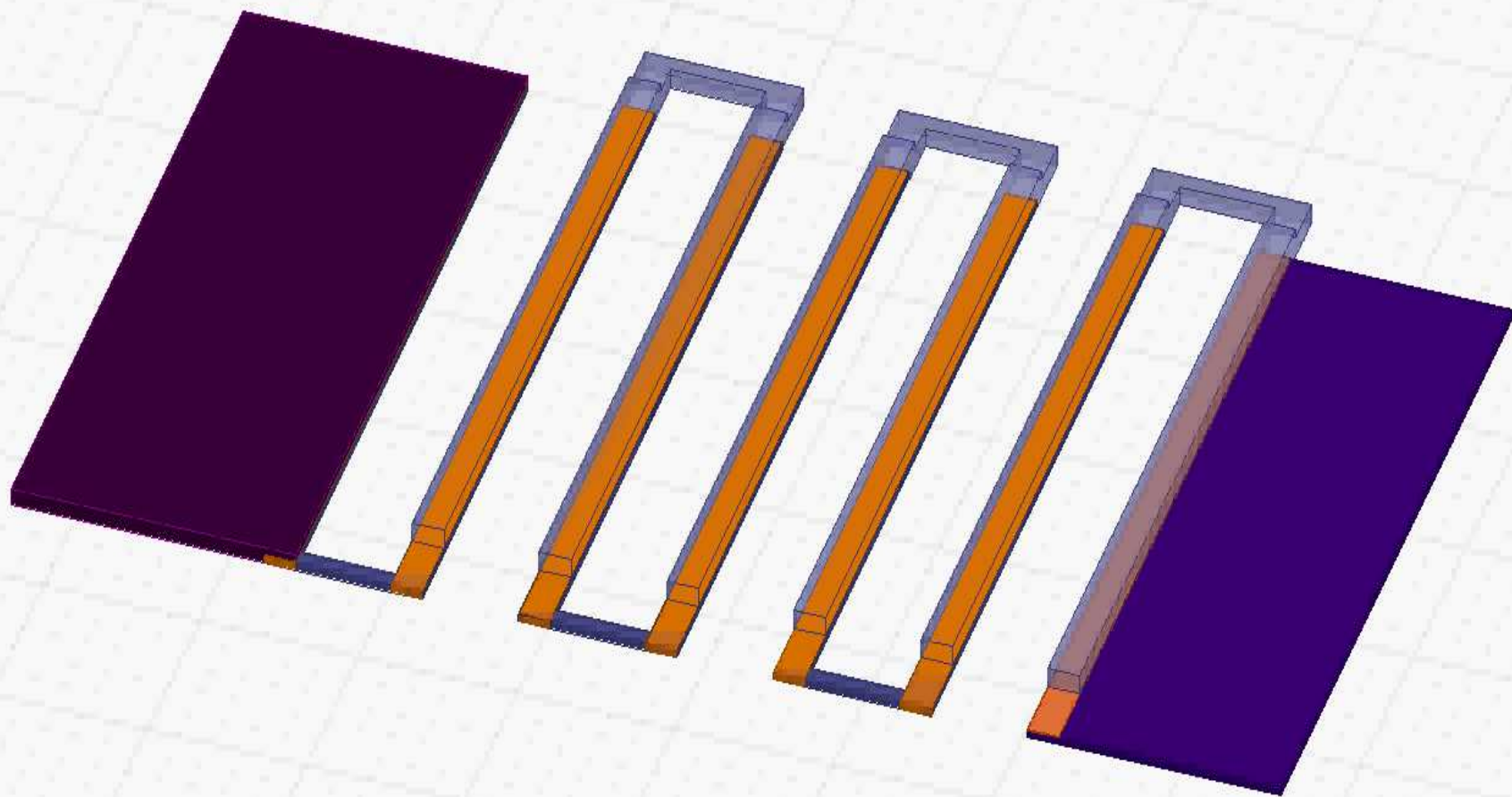
- To minimize parasitic capacitance to ground in Josephson junction arrays use:
 - Deeper junction (more rectangular, less square)
 - Larger junctions
 - Thicker second layer
 - There is a lower bound on the parasitic capacitance
 - Capacitance to ground varies linearly with island separation

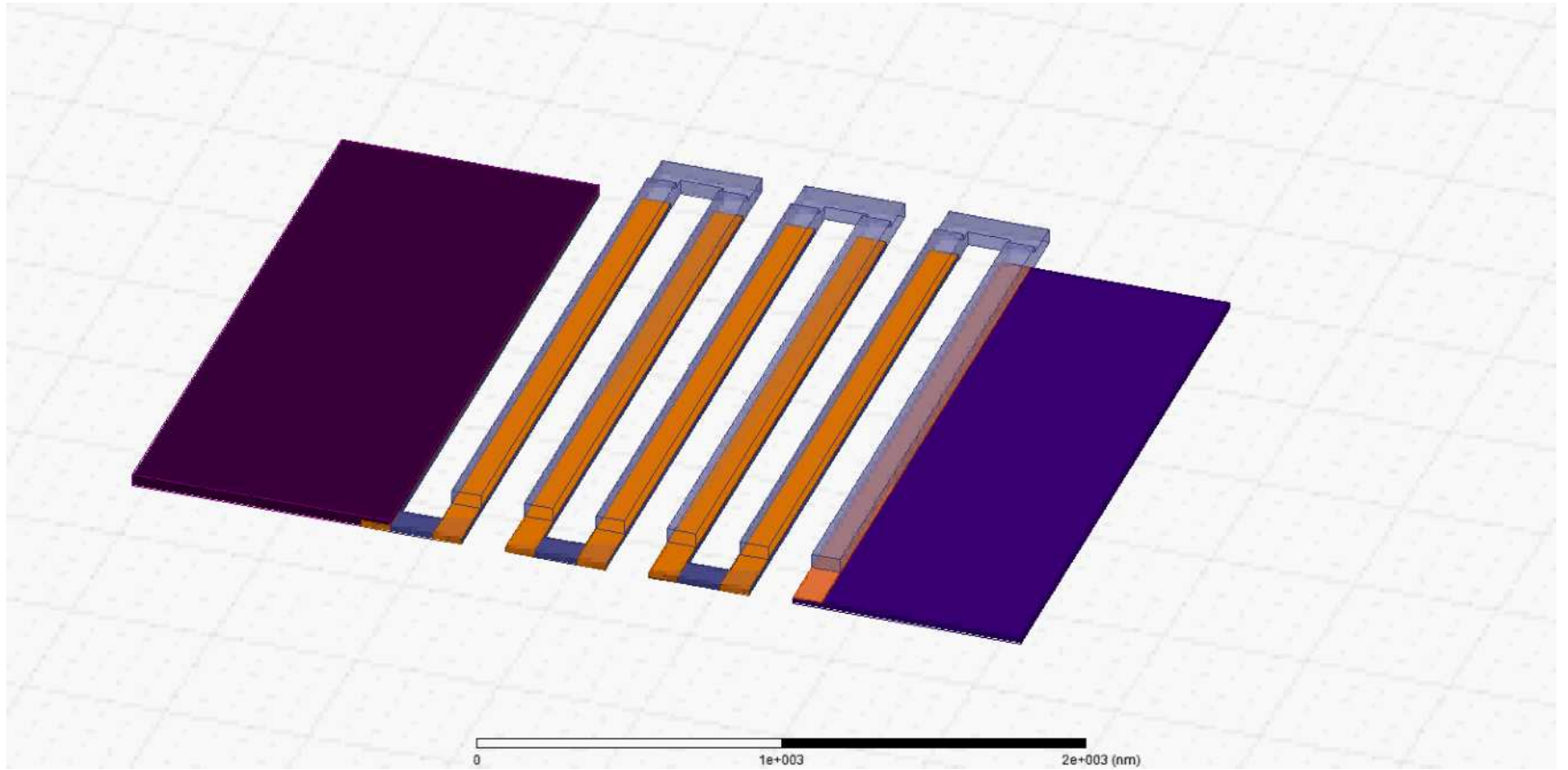
Bridge Free

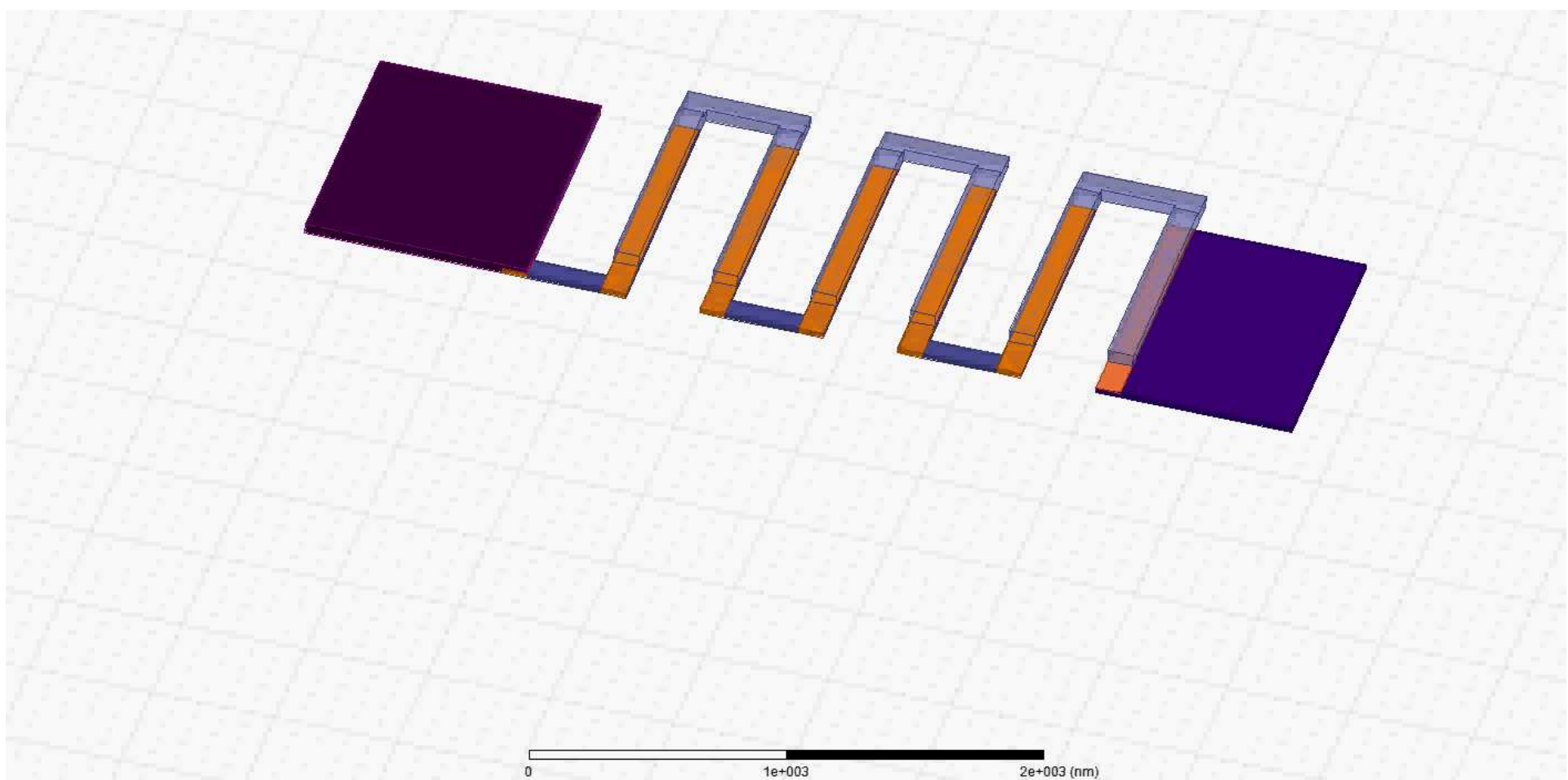






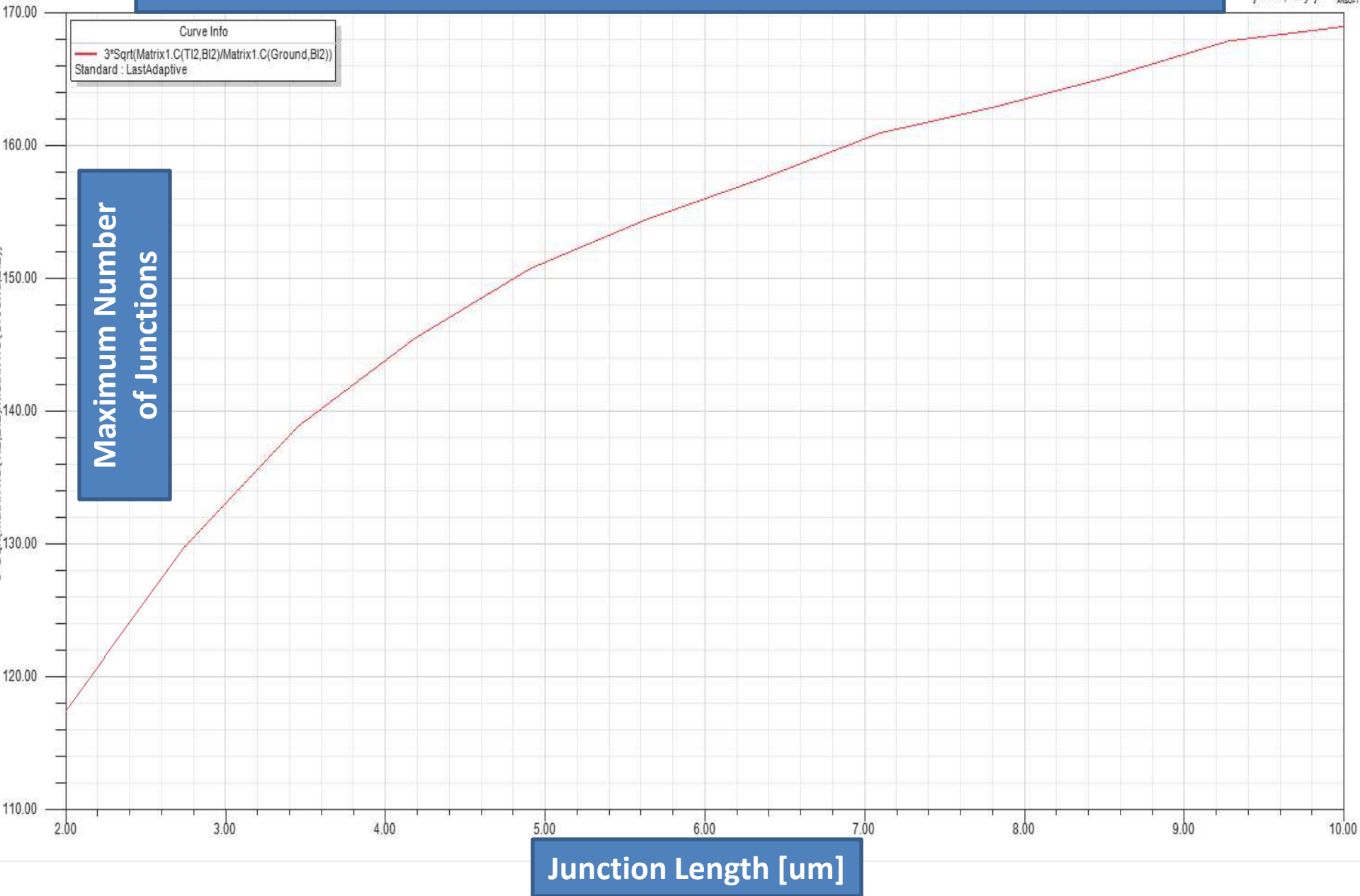






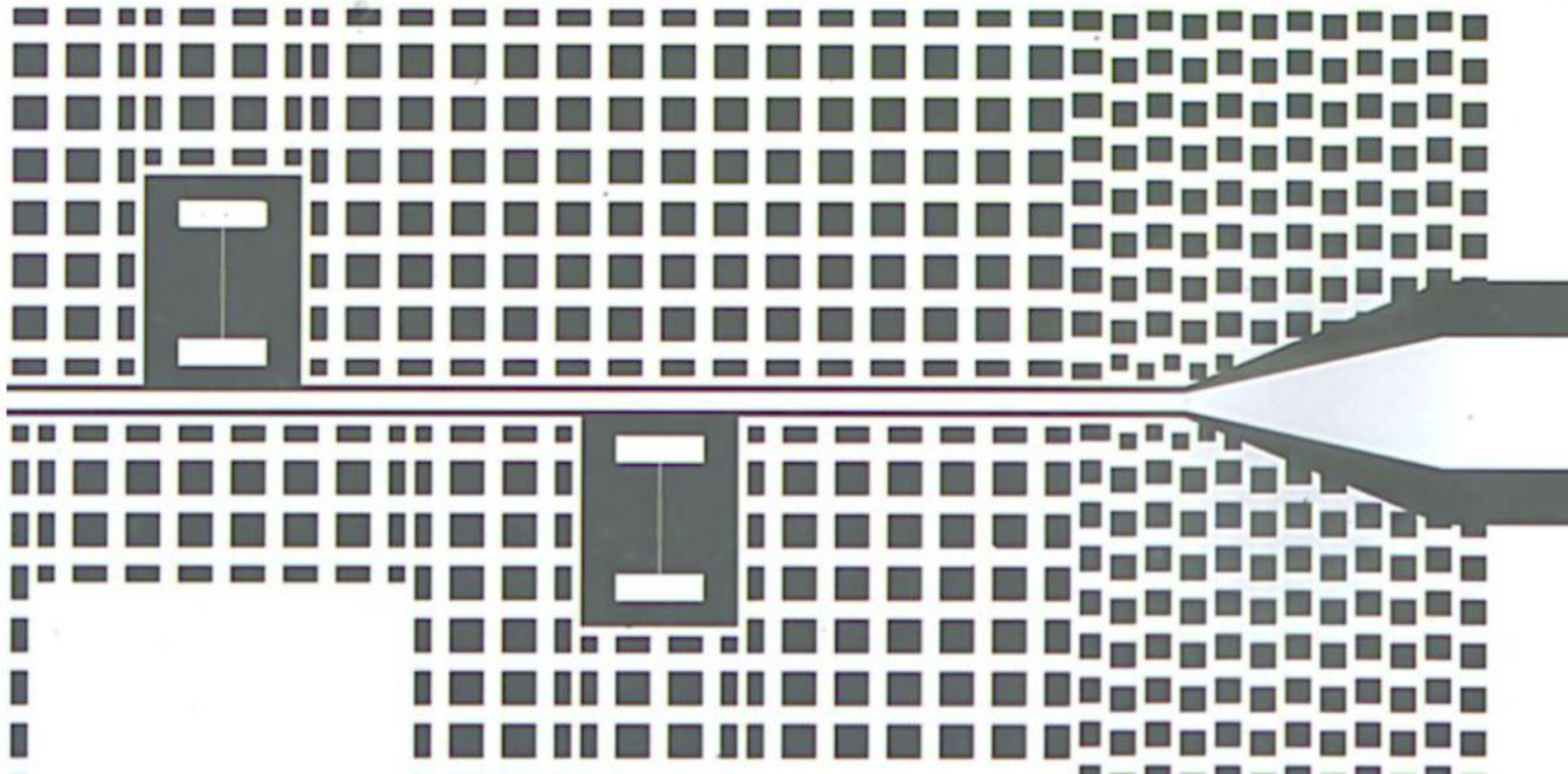
Maximum Number of Junctions for Junction Width of 100nm

jw100_vary jD



Junction Array Q

Junction Array Loss



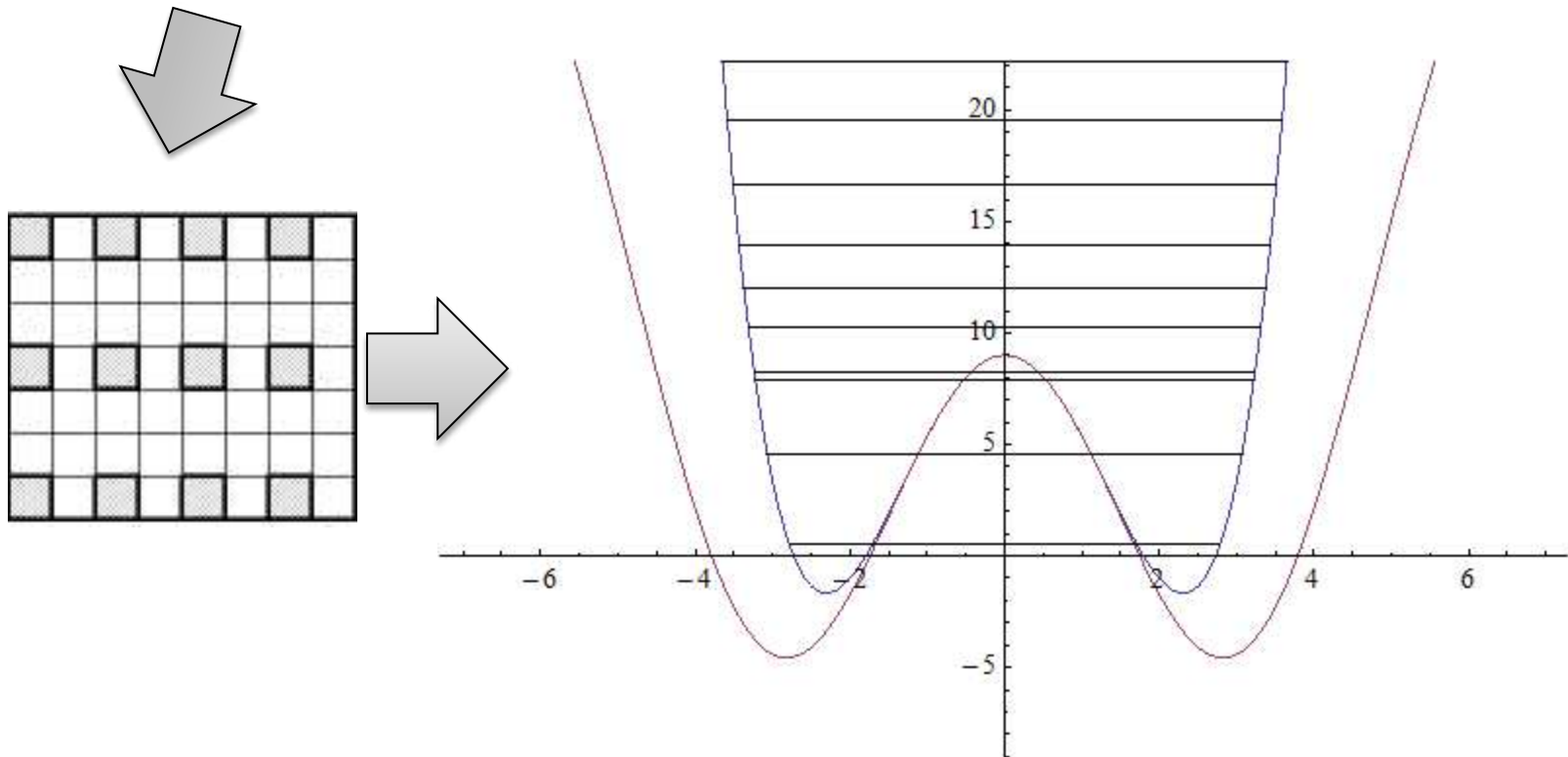
		W													
		Resonator C			Top Ground			Bottom C-Pin			Bottom Grnd				
S	L	10	20	30	10	20	30	10	20	30	10	20	30	L	S
60	10	0.05	0.11	0.21	2.21	3.21	3.98	0.75	0.86	0.95	3.01	3.67	4.32	10	60
	30	0.14	0.30	0.53	4.08	5.40	6.51	1.11	1.30	1.43	4.46	5.58	6.47	30	
	50	0.26	0.49	0.88	5.68	7.31	8.58	1.49	1.64	1.82	5.99	7.04	8.28	50	
	70	0.39	0.74	1.25	7.26	9.13	10.56	1.78	2.01	2.17	7.30	8.78	10.02	70	
	90	0.52	0.94	1.59	8.73	10.85	12.42	2.05	2.25	2.43	8.62	10.06	11.52	90	
	110	0.64	1.18	1.94	10.28	12.69	14.45	2.27	2.54	2.72	9.83	11.80	13.35	110	
	120	0.69	1.28	2.15	11.01	13.66	15.63	2.34	2.63	2.85	10.33	12.55	14.50	120	
50	10	0.04	0.07	0.14	2.25	3.23	4.08	0.83	0.92	1.03	2.82	3.31	4.03	10	50
	30	0.11	0.20	0.37	4.13	5.57	6.73	1.34	1.46	1.64	4.50	5.25	6.36	30	
	50	0.18	0.35	0.60	5.85	7.50	8.87	1.70	1.88	2.06	5.69	6.81	7.99	50	
	70	0.28	0.50	0.86	7.44	9.38	10.97	2.10	2.25	2.48	7.16	8.19	9.71	70	
	90	0.37	0.67	1.12	8.94	11.24	13.03	2.43	2.65	2.89	8.46	9.96	11.50	90	
	110	0.46	0.85	1.37	10.56	13.23	15.12	2.67	3.03	3.21	9.58	11.77	13.31	110	
	120	0.49	0.90	1.48	11.31	14.10	16.17	2.75	3.09	3.35	10.03	12.27	14.22	120	
40	10	0.02	0.06	0.10	2.26	3.28	4.18	0.77	1.10	1.19	2.04	3.37	3.93	10	40
	30	0.06	0.14	0.25	4.25	5.73	6.89	1.36	1.71	1.86	3.64	5.08	6.03	30	
	50	0.12	0.25	0.42	5.99	7.74	9.20	1.87	2.25	2.41	5.00	6.76	7.79	50	
	70	0.18	0.36	0.60	7.58	9.68	11.33	2.32	2.68	2.90	6.31	8.06	9.38	70	
	90	0.26	0.48	0.80	9.30	11.62	13.50	2.90	3.13	3.40	8.06	9.55	11.23	90	
	110	0.33	0.59	0.97	10.90	13.61	15.69	3.22	3.55	3.79	9.27	11.20	12.93	110	
	120	0.36	0.65	1.07	11.74	14.62	16.85	3.38	3.72	4.02	9.93	11.99	14.06	120	
30	10	0.01	0.04	0.07	2.29	3.40	4.25	0.83	1.26	1.38	1.70	3.04	3.68	10	30
	30	0.04	0.10	0.17	4.32	5.87	7.06	1.53	2.01	2.18	3.13	4.76	5.64	30	
	50	0.08	0.17	0.30	6.15	8.01	9.47	2.21	2.69	2.88	4.56	6.34	7.47	50	
	70	0.12	0.26	0.43	7.84	10.03	11.80	2.73	3.28	3.52	5.71	7.84	9.18	70	
	90	0.17	0.35	0.57	9.63	12.13	14.13	3.34	3.86	4.13	7.17	9.36	10.91	90	
	110	0.23	0.43	0.70	11.31	14.09	16.46	3.87	4.32	4.65	8.58	10.75	12.68	110	
	120	0.25	0.47	0.77	12.18	15.25	17.69	4.08	4.59	4.95	9.21	11.62	13.79	120	
20	10	0.01	0.02	0.05	2.38	3.44	4.42	1.01	1.37	1.74	1.51	2.39	3.65	10	20
	30	0.03	0.06	0.13	4.52	6.07	7.37	1.91	2.36	2.75	2.85	4.11	5.59	30	
	50	0.05	0.12	0.22	6.45	8.40	9.95	2.70	3.29	3.67	4.05	5.80	7.41	50	
	70	0.08	0.17	0.31	8.29	10.59	12.46	3.45	4.06	4.47	5.25	7.18	8.99	70	

Fluxonium and Beyond

From Scratch = Faster Diagonalization

$$\tilde{H} = \tilde{E}_c \tilde{q}^2 + \tilde{E}_L \tilde{\phi}^2 - \tilde{E}_J \text{Cos}[\tilde{\phi} - \tilde{\phi}_e]$$

Turn analytic problem into algebraic problem



Non-Commutative Algebra in Mathematica



```
(*break up cosine *)
cosList[N_] := ExpandXPower @ (List@@ (Cos[x] // Series[#, {x, 0, N}] & // Normal))
cos = cosList[5] /. x -> (aL + aR) // ExpandNCM
(*give me taylor expanded cos up to 5th order and then x replaced by aL +
aR then fully power expanded by noncommutative algebra*)
```

$$\left\{1, \frac{1}{2} (-aL ** aL - aL ** aR - aR ** aL - aR ** aR), \right. \\ \left. \frac{1}{24} (aL ** aL ** aL ** aL + aL ** aL ** aL ** aR + aL ** aL ** aR ** aL + aL ** aL ** aR ** aR + \right. \\ aL ** aR ** aL ** aL + aL ** aR ** aL ** aR + aL ** aR ** aR ** aL + aL ** aR ** aR ** aR + \\ aR ** aL ** aL ** aL + aR ** aL ** aL ** aR + aR ** aL ** aR ** aL + aR ** aL ** aR ** aR + \\ \left. aR ** aR ** aL ** aL + aR ** aR ** aL ** aR + aR ** aR ** aR ** aL + aR ** aR ** aR ** aR) \right\}$$

```

CL = 5;
BL = 100;
ξ = √(1/2 * √(Ec/E1)) /. {Ec → 2.5, E1 → 0.5};
ϕe = π;
T
cosList[N_] := ExpandXPower @ (List@@ (Cos[ξ x] // Series[#, {x, 0, N}] & // Normal))
cos0 = cosList[CL] /. x → (aL + aR);
cos1 = cos0 // ExpandNCM
(*CSElIJ[i_, j_] := Plus@@(bra[i]**#&/@ ((#**ket[j]&/@ cos1)//ExpandNCM)) *)
CSElIJ[i_, j_] := Evaluate[
  cos2 = (#**ket[j] &/@ cos1) // ExpandNCM;
  cos2 = (bra[i]**# &/@ cos2) // ExpandNCM;
  Plus@@cos2
];
sinList[N_] := ExpandXPower @ (List@@ (Sin[ξ x] // Series[#, {x, 0, N}] & // Normal))
sin0 = sinList[CL] /. x → (aL + aR);
sin1 = sin0 // ExpandNCM;
SINelIJ[i_, j_] := Evaluate[
  sin2 = (#**ket[j] &/@ sin1) // ExpandNCM;
  sin2 = (bra[i]**# &/@ sin2) // ExpandNCM;
  Plus@@sin2
];
(*can keep the numbers factored out might run faster *)
HoMtrx = Table[Cos[ϕe] CSElIJ[n, j] - Sin[ϕe] * SINelIJ[n, j], {n, 0, BL}, {j, 0, BL}];
HoMtrx // MatrixForm // TraditionalForm
evals = Eigenvalues[HoMtrx];

```

$-\frac{1}{2\sqrt{2}}$	0	$\frac{1}{2\sqrt{6}}$	0	
0	$\frac{5}{2\sqrt{6}} - \sqrt{\frac{3}{2}}$	0	$\frac{\sqrt{\frac{5}{6}}}{2}$	
$\frac{1}{8}$	0	$\frac{7}{2\sqrt{3}} - \sqrt{3}$	0	
0	$\frac{5}{8}$	0	$\frac{\sqrt{5}}{2}$	
$\frac{7}{2\sqrt{3}} - \sqrt{3}$	0	$\frac{13}{8}$	0	$\frac{11\sqrt{2}}{2}$
0	$\frac{\sqrt{5}}{2}$	0	$\frac{25}{8}$	
$\frac{\sqrt{\frac{5}{2}}}{2}$	0	$\frac{11\sqrt{\frac{5}{6}}}{2} - \sqrt{\frac{15}{2}}$	0	
0	$\frac{\sqrt{\frac{35}{6}}}{2}$	0	$\frac{13\sqrt{\frac{7}{6}}}{2} - \sqrt{\frac{21}{2}}$	
0	0	$\frac{\sqrt{\frac{35}{3}}}{2}$	0	$5\sqrt{2}$
0	0	0	$\frac{\sqrt{21}}{2}$	
0	0	0	0	

From Scratch = Faster Diagonalization

$$\tilde{H} = \tilde{E}_c \tilde{q}^2 + \tilde{E}_L \tilde{\phi}^2 - \tilde{E}_J \text{Cos}[\tilde{\phi} - \tilde{\phi}_e]$$

- Turn analytic problem into algebraic problem

